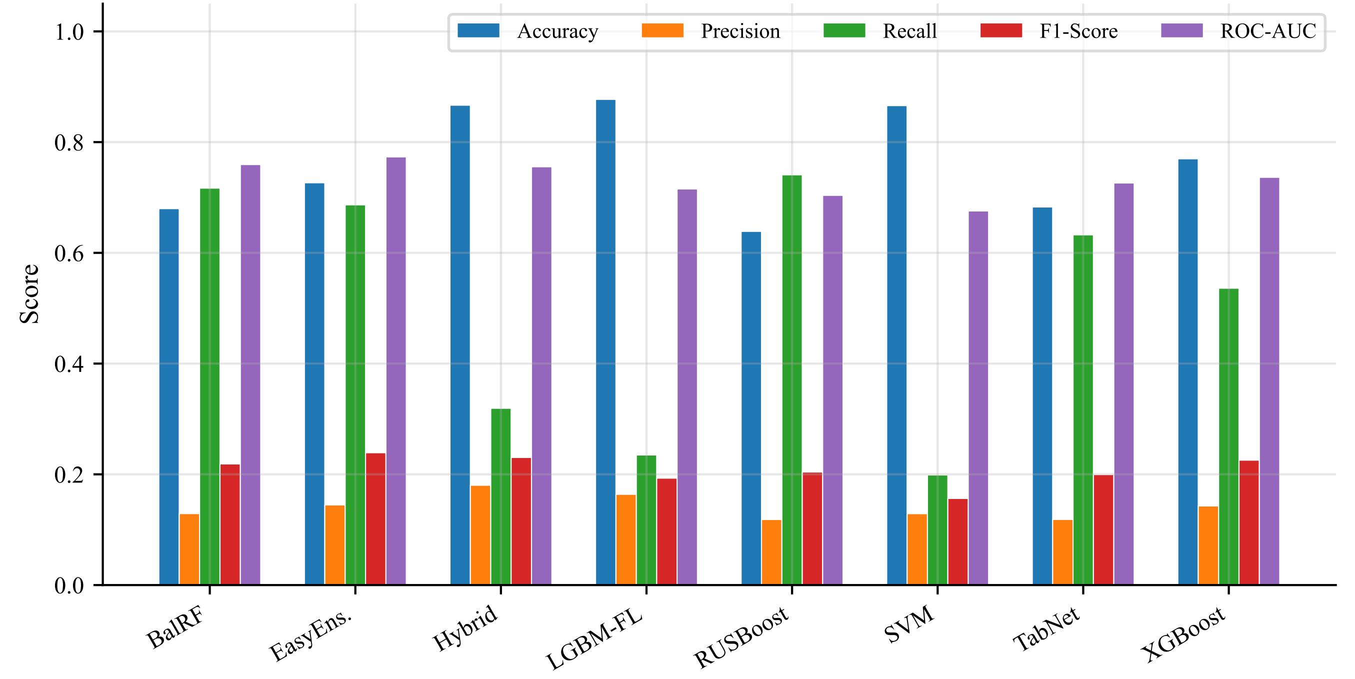
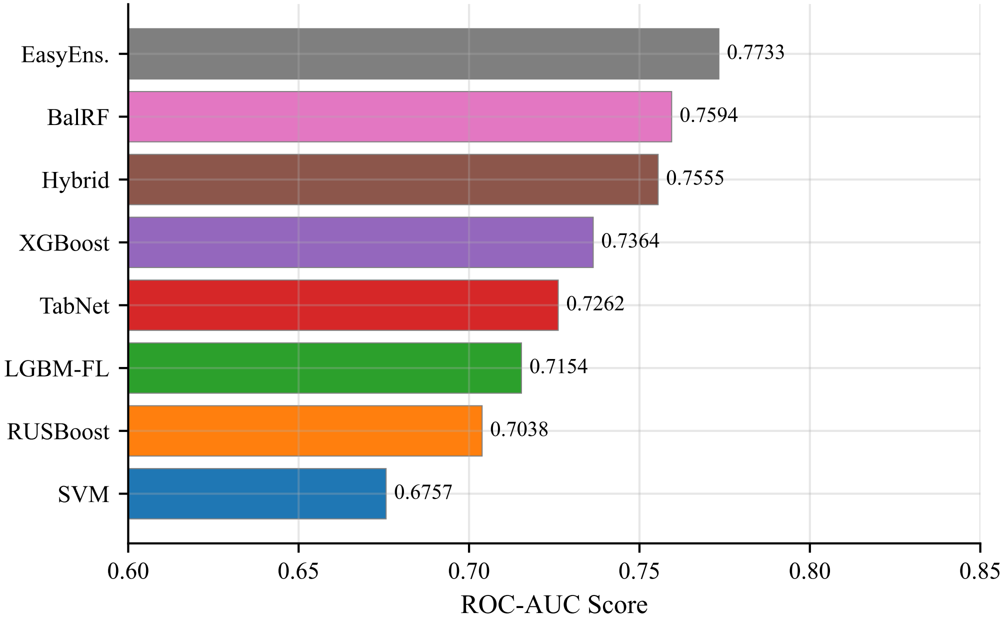

Accuracy
Precision
Recall
F1-Score
ROC-AUC
Score
1.0
0.8
0.6
0.4
0.2
0.0
BalRF
EasyEns.
Hybrid
LGBM-FL
RUSBoost
SVM
TabNet
XGBoost


EasyEns.
BalRF
Hybrid
XGBoost
TabNet
LGBM-FL
RUSBoost
SVM
0.7733
0.7594
0.7555
0.7364
0.7262
0.7154
0.7038
0.6757
0.60
0.65
0.70
0.75
0.80
0.85
ROC-AUC Score

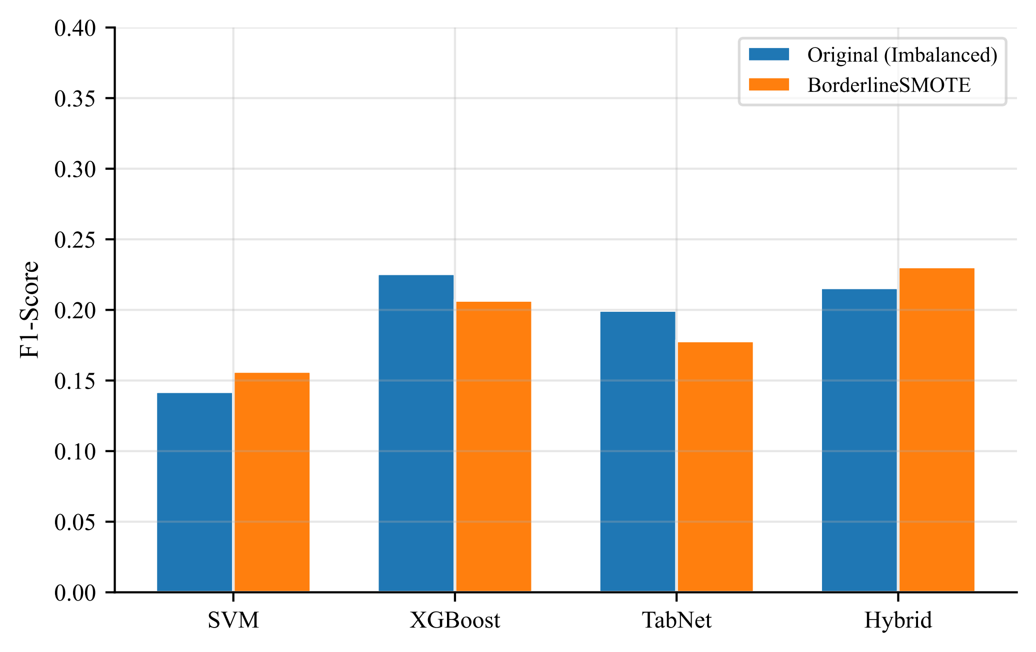
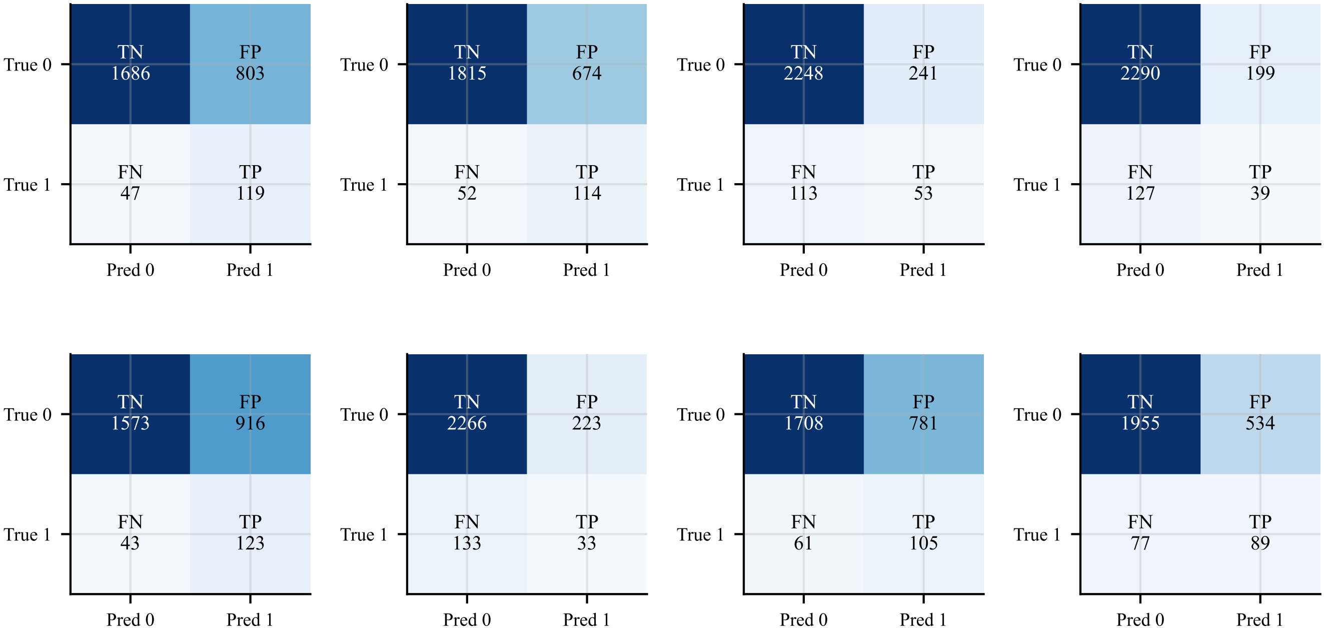
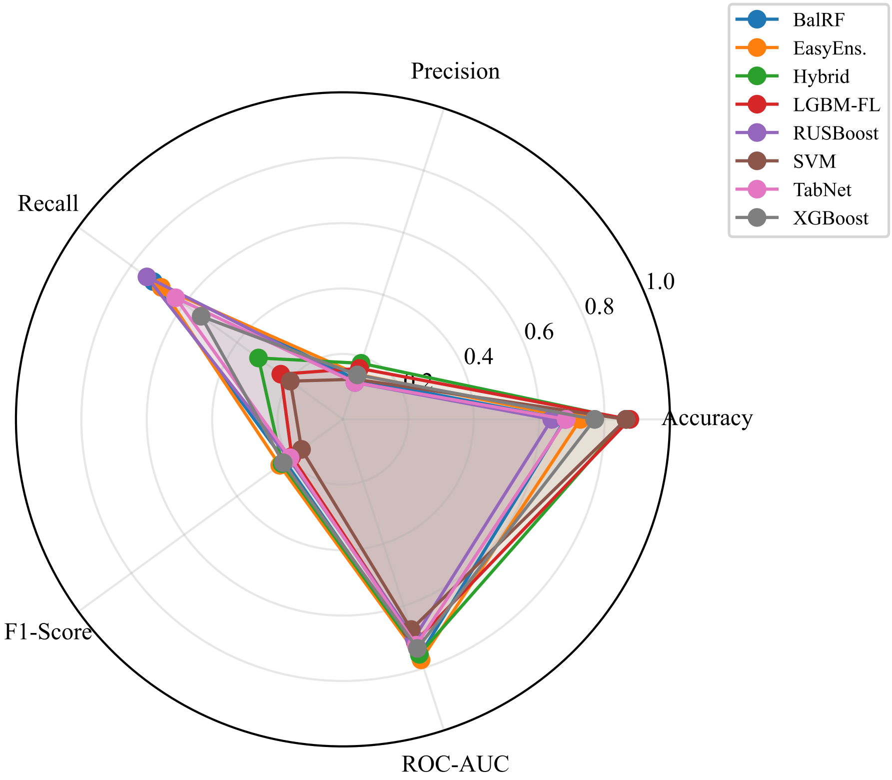
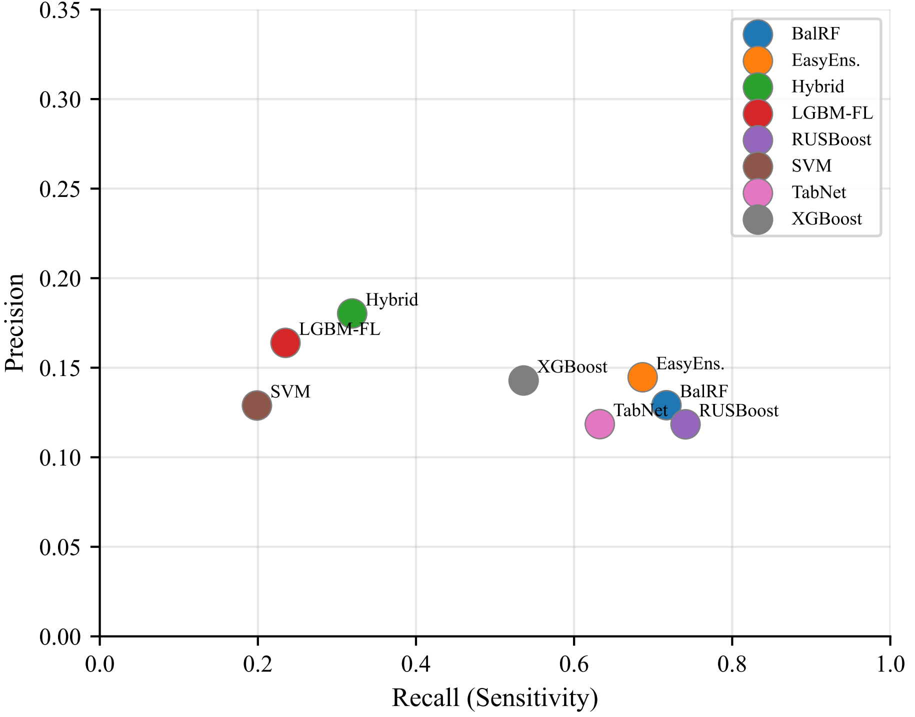

0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
F1-Score
Original (Imbalanced)
BorderlineSMOTE
SVM
XGBoost
TabNet
Hybrid

Fig. 4 — Confusion Matrices (Best Strategy per Model)

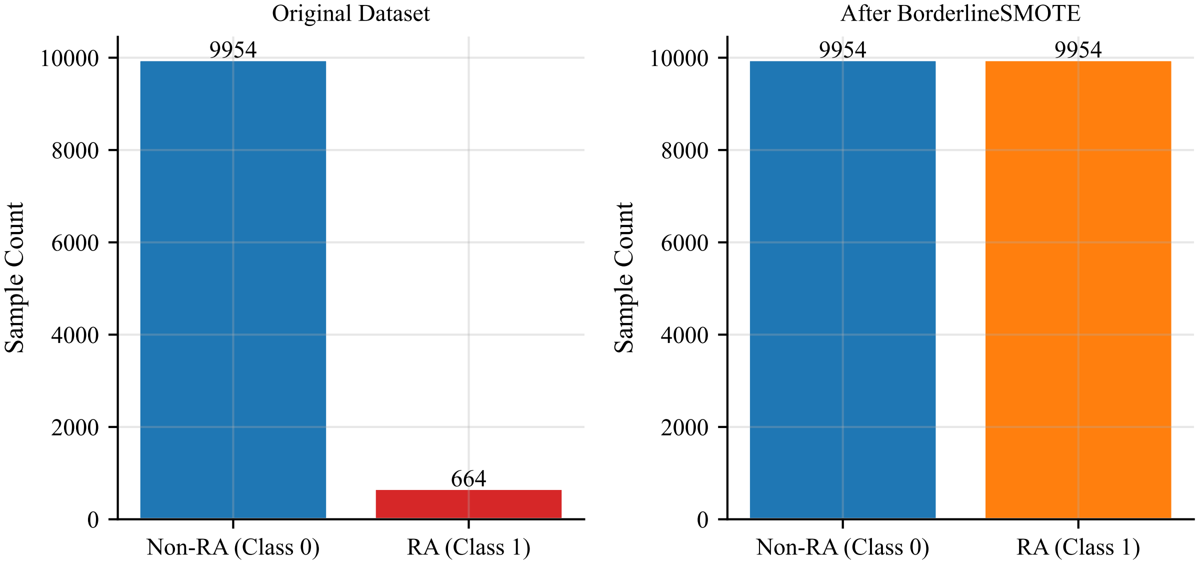




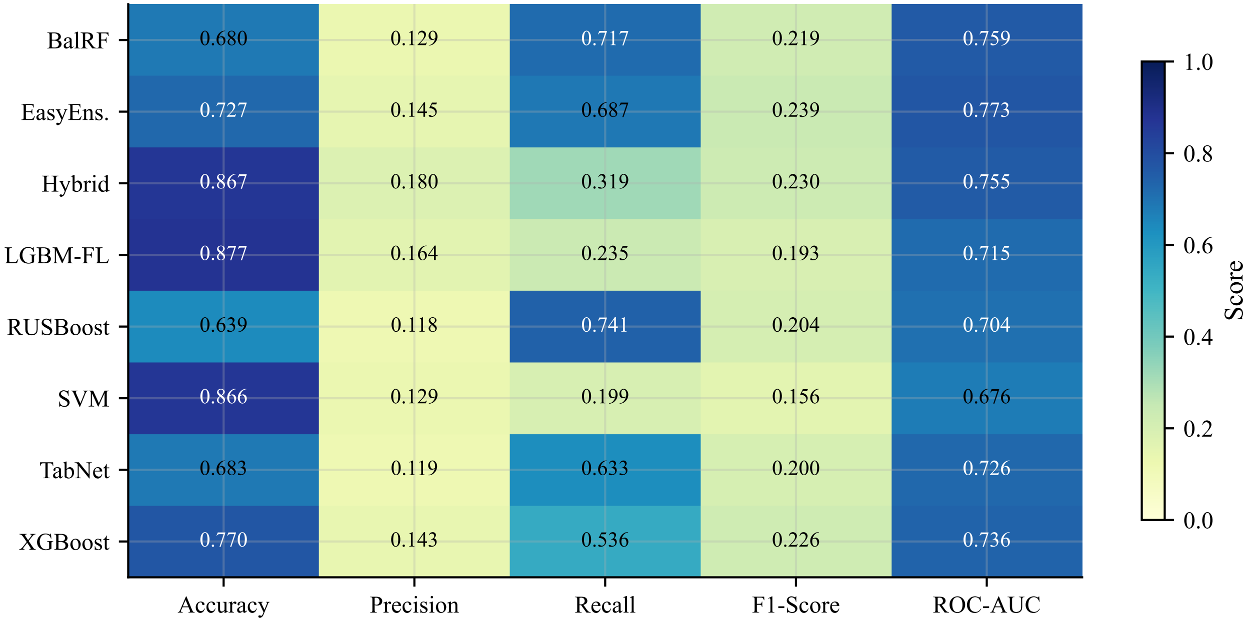
Precision
Recall
F1-Score
ROC-AUC
Accuracy
0.2
0.4
0.6
0.8
1.0
BalRF
EasyEns.
Hybrid
LGBM-FL
RUSBoost
SVM
TabNet
XGBoost

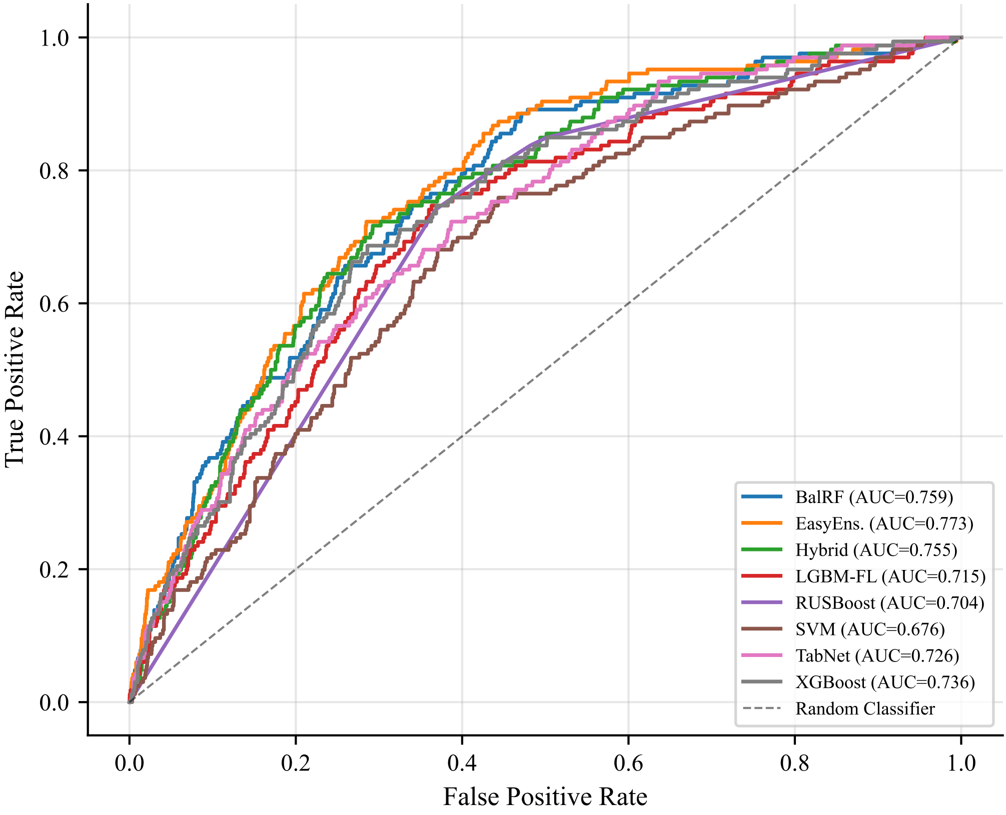


0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
Precision
0.0
0.2
0.4
0.6
0.8
1.0
Recall (Sensitivity)
Hybrid
LGBM-FL
SVM
XGBoost
EasyEns.
BalRF
TabNet
RUSBoost
BalRF
EasyEns.
Hybrid
LGBM-FL
RUSBoost
SVM
TabNet
XGBoost

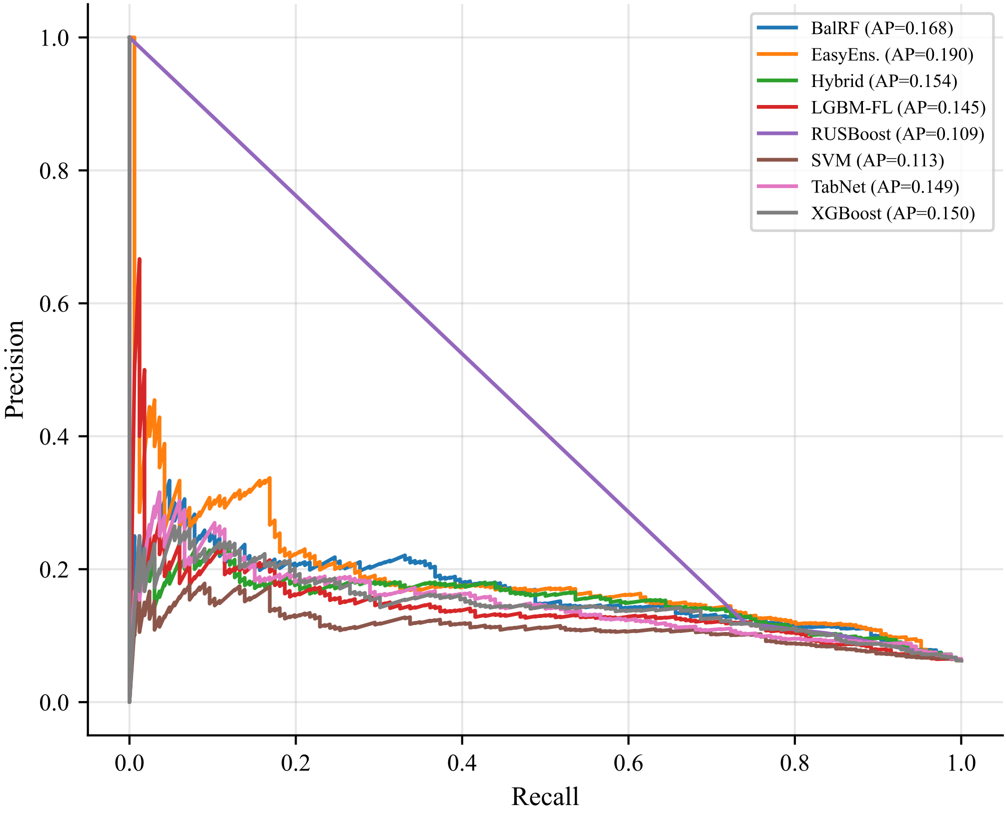


Original Dataset
After BorderlineSMOTE
Sample Count
0
2000
4000
6000
8000
10000
9954
664
9954
9954
Non-RA (Class 0)
RA (Class 1)
Non-RA (Class 0)
RA (Class 1)

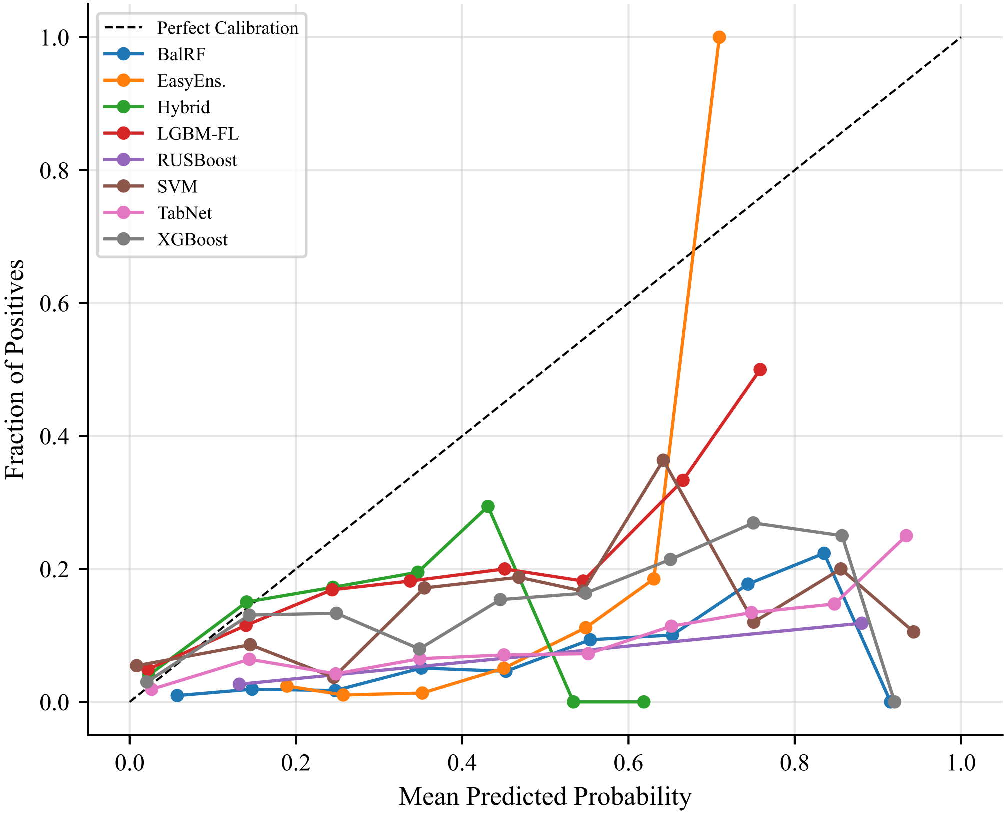

BalRF
EasyEns.
Hybrid
LGBM-FL
RUSBoost
SVM
TabNet
XGBoost
Accuracy
Precision
Recall
F1-Score
ROC-AUC
0.680
0.129
0.717
0.219
0.759
0.727
0.145
0.687
0.239
0.773
0.867
0.180
0.319
0.230
0.755
0.877
0.164
0.235
0.193
0.715
0.639
0.118
0.741
0.204
0.704
0.866
0.129
0.199
0.156
0.676
0.683
0.119
0.633
0.200
0.726
0.770
0.143
0.536
0.226
0.736
Score
1.0
0.8
0.6
0.4
0.2
0.0



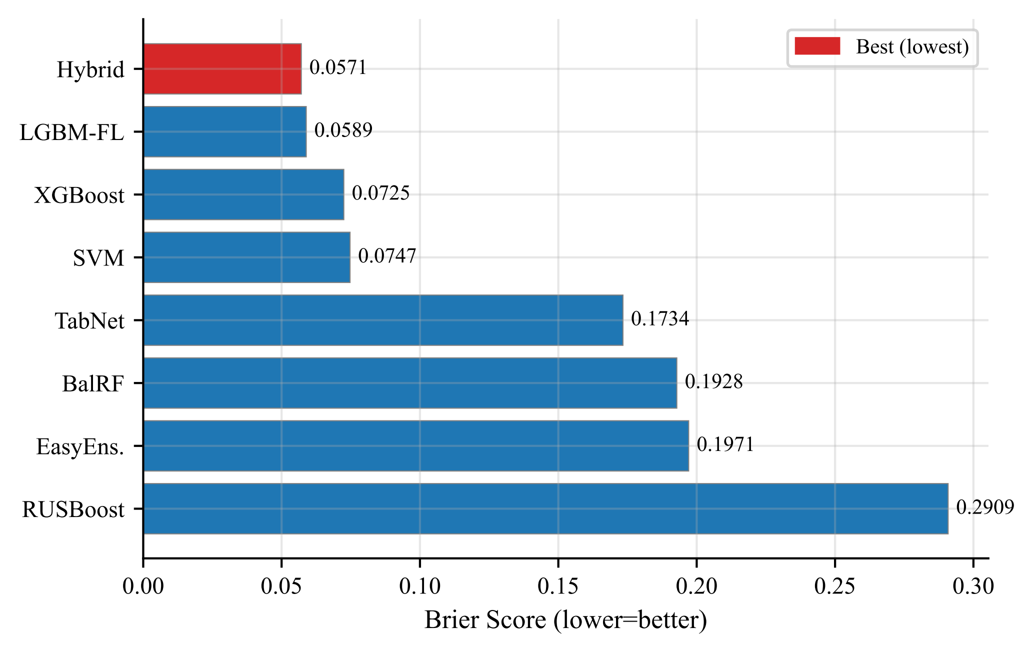
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
0.0
0.2
0.4
0.6
0.8
1.0
BalRF (AUC=0.759)
EasyEns. (AUC=0.773)
Hybrid (AUC=0.755)
LGBM-FL (AUC=0.715)
RUSBoost (AUC=0.704)
SVM (AUC=0.676)
TabNet (AUC=0.726)
XGBoost (AUC=0.736)
Random Classifier



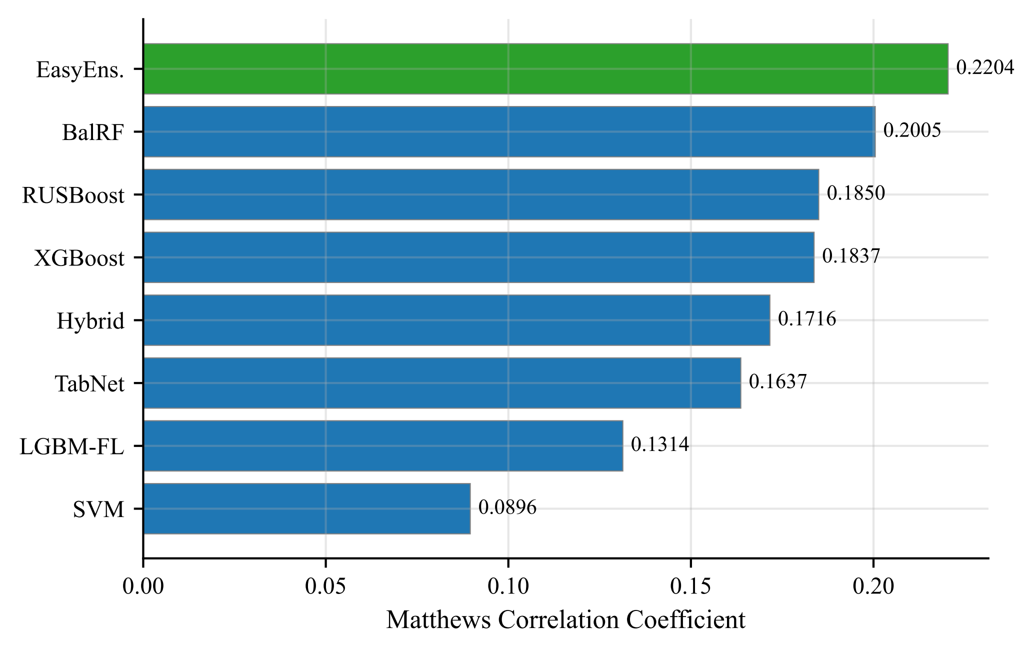
BalRF (AP=0.168)
EasyEns. (AP=0.190)
Hybrid (AP=0.154)
LGBM-FL (AP=0.145)
RUSBoost (AP=0.109)
SVM (AP=0.113)
TabNet (AP=0.149)
XGBoost (AP=0.150)
Precision
Recall
0.0
0.2
0.4
0.6
0.8
1.0



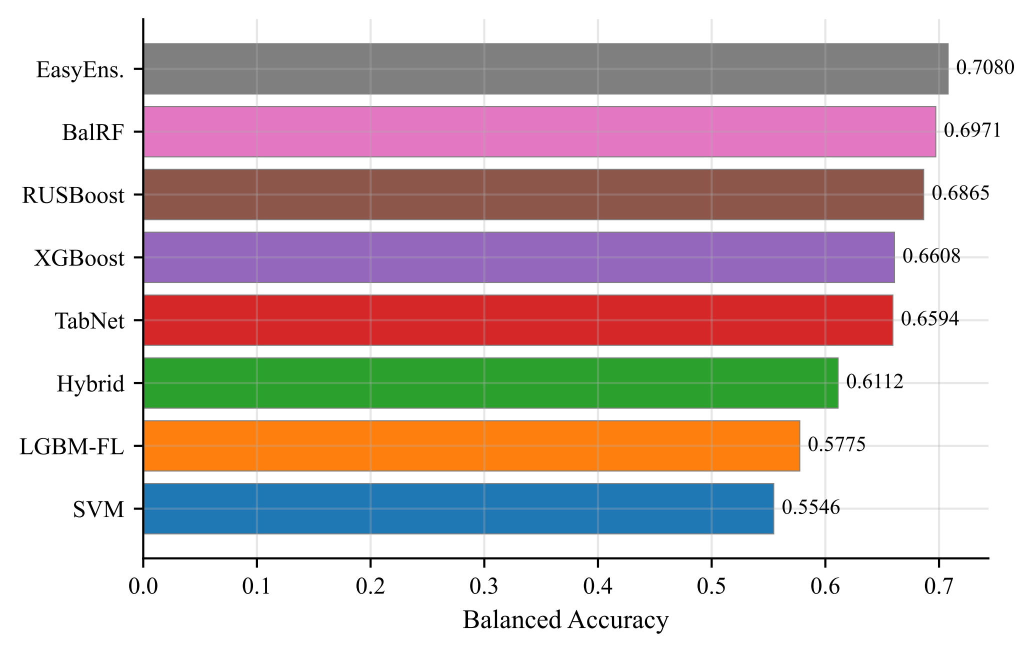
Perfect Calibration
BalRF
EasyEns.
Hybrid
LGBM-FL
RUSBoost
SVM
TabNet
XGBoost
Fraction of Positives
Mean Predicted Probability
0.0
0.2
0.4
0.6
0.8
1.0

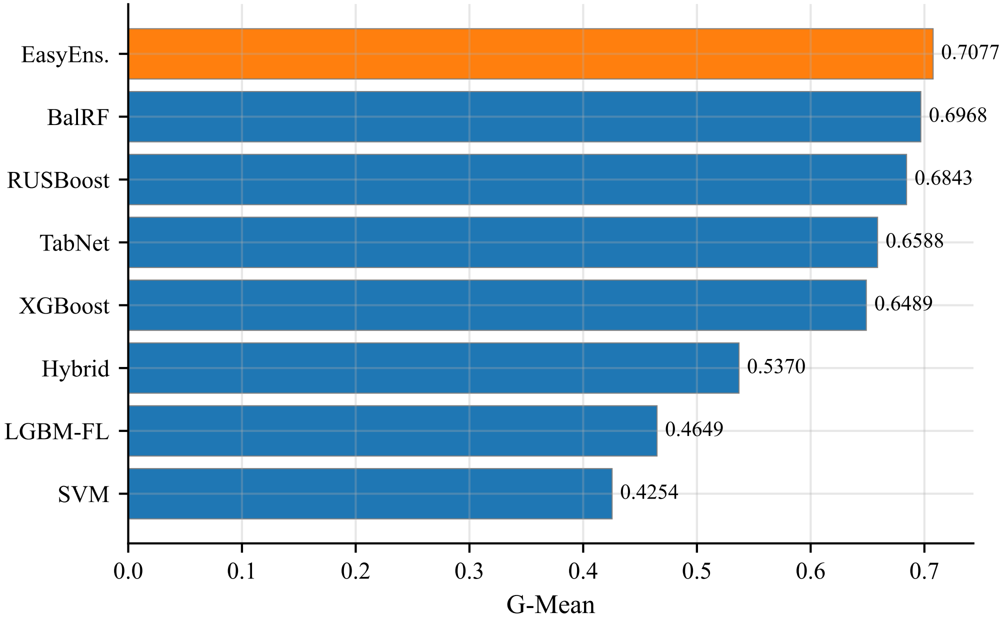


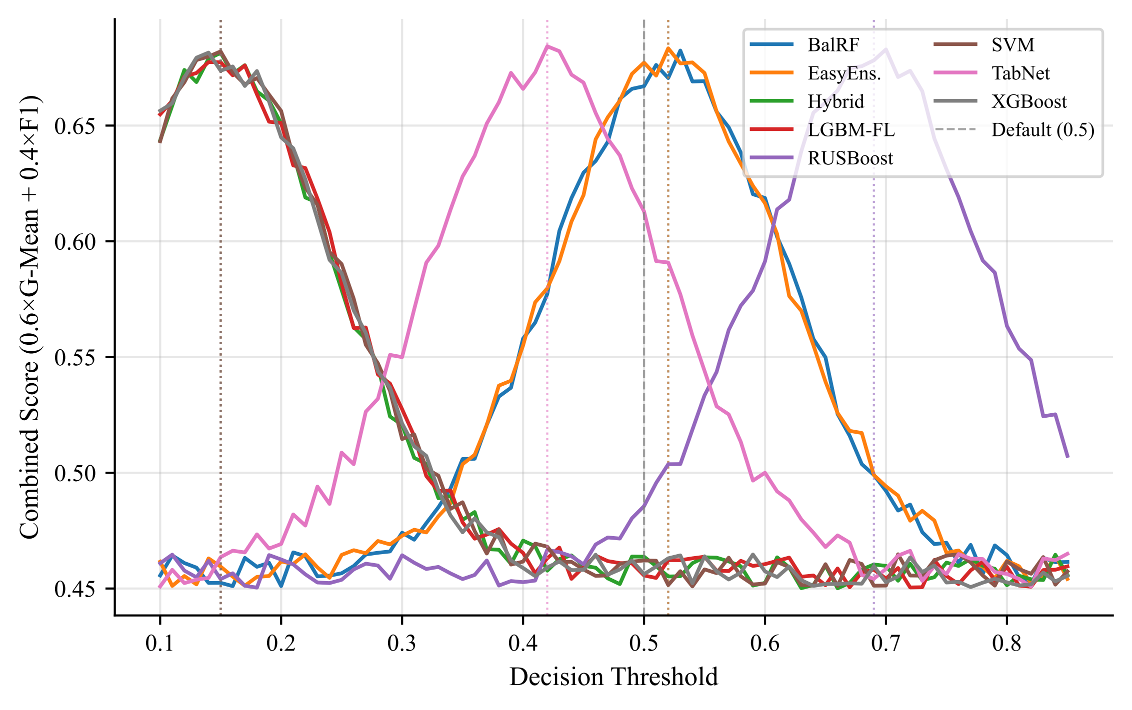
Best (lowest)
Hybrid
LGBM-FL
XGBoost
SVM
TabNet
BalRF
EasyEns.
RUSBoost
0.0571
0.0589
0.0725
0.0747
0.1734
0.1928
0.1971
0.2909
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Brier Score (lower=better)



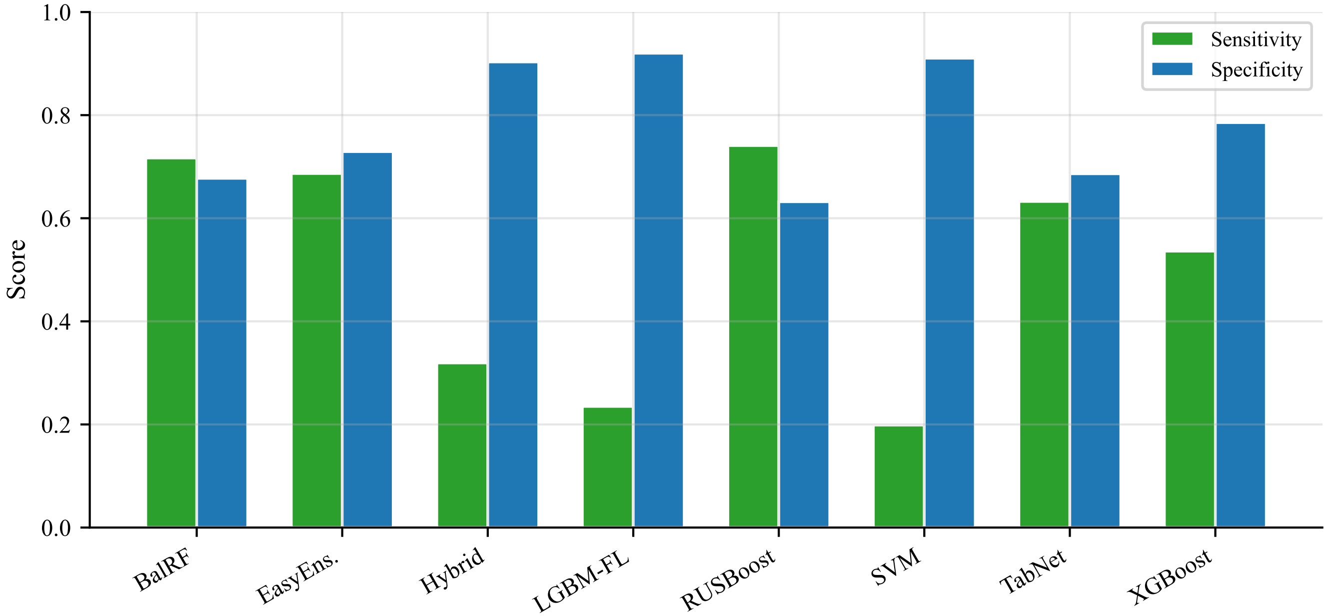
EasyEns.
BalRF
RUSBoost
XGBoost
Hybrid
TabNet
LGBM-FL
SVM
0.2204
0.2005
0.1850
0.1837
0.1716
0.1637
0.1314
0.0896
0.00
0.05
0.10
0.15
0.20
Matthews Correlation Coefficient


EasyEns.
BalRF
RUSBoost
XGBoost
TabNet
Hybrid
LGBM-FL
SVM
0.7080
0.6971
0.6865
0.6608
0.6594
0.6112
0.5775
0.5546
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Balanced Accuracy

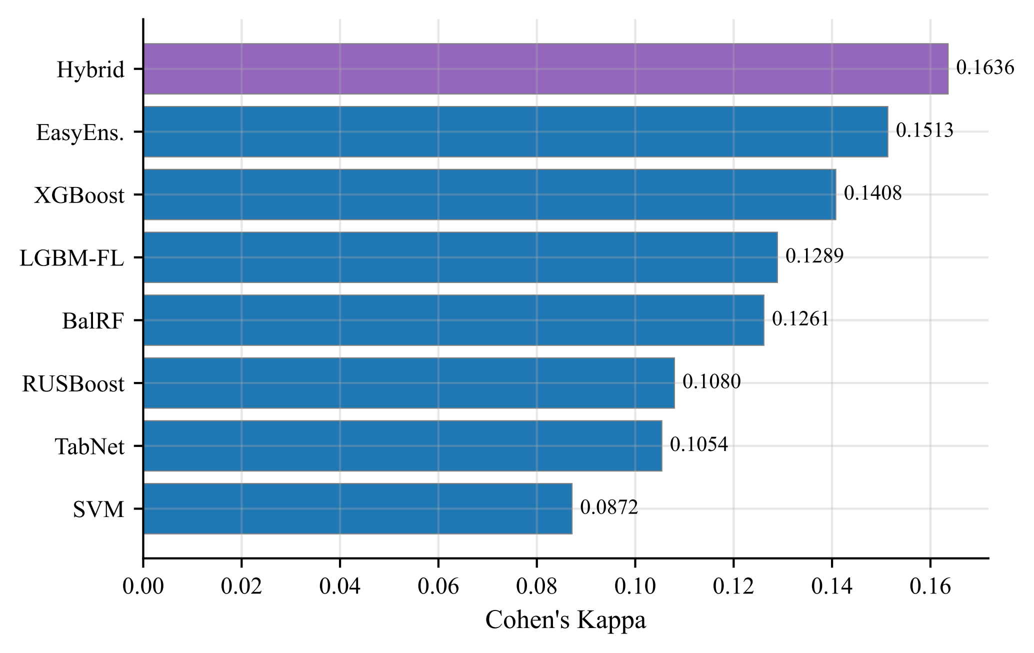

EasyEns.
BalRF
RUSBoost
TabNet
XGBoost
Hybrid
LGBM-FL
SVM
0.7077
0.6968
0.6843
0.6588
0.6489
0.5370
0.4649
0.4254
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
G-Mean

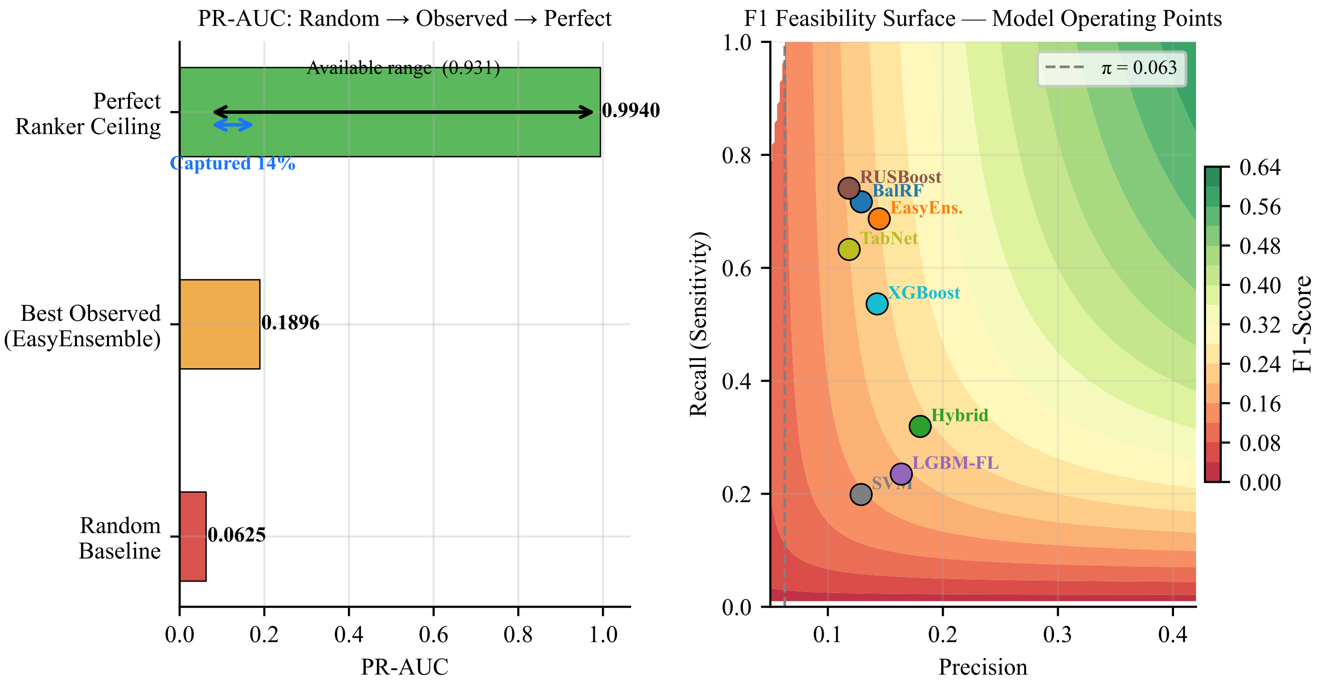


Combined Score (0.6×G-Mean + 0.4×F1)
Decision Threshold
0.45
0.50
0.55
0.60
0.65
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
BalRF
EasyEns.
Hybrid
LGBM-FL
RUSBoost
SVM
TabNet
XGBoost
Default (0.5)

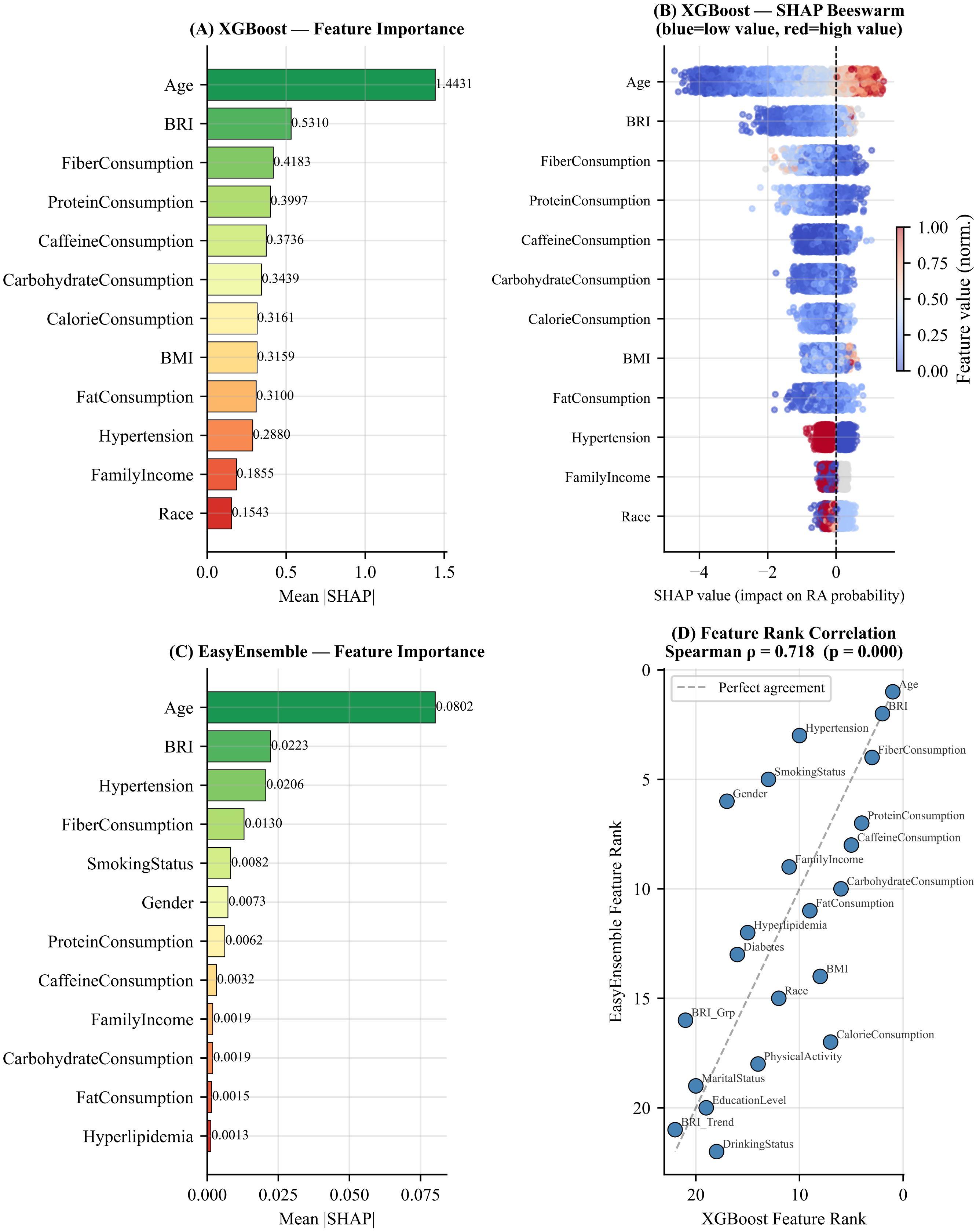


Sensitivity
Specificity
Score
1.0
0.8
0.6
0.4
0.2
0.0
BalRF
EasyEns.
Hybrid
LGBM-FL
RUSBoost
SVM
TabNet
XGBoost


Hybrid
EasyEns.
XGBoost
LGBM-FL
BalRF
RUSBoost
TabNet
SVM
0.1636
0.1513
0.1408
0.1289
0.1261
0.1080
0.1054
0.0872
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
Cohen's Kappa



PR-AUC: Random → Observed → Perfect
Available range (0.931)
Perfect
Ranker Ceiling
0.9940
Captured 14%
Best Observed
(EasyEnsemble)
0.1896
Random
Baseline
0.0625
0.0
0.2
0.4
0.6
0.8
1.0
PR-AUC
F1 Feasibility Surface — Model Operating Points
π = 0.063
RUSBoost
BalRF
EasyEns.
TabNet
XGBoost
Hybrid
LGBM-FL
SVM
Recall (Sensitivity)
Precision
F1-Score
0.00
0.08
0.16
0.24
0.32
0.40
0.48
0.56
0.64


(A) XGBoost — Feature Importance
Age 1.4431
BRI 0.5310
FiberConsumption 0.4183
ProteinConsumption 0.3997
CaffeineConsumption 0.3736
CarbohydrateConsumption 0.3439
CalorieConsumption 0.3161
BMI 0.3159
FatConsumption 0.3100
Hypertension 0.2880
FamilyIncome 0.1855
Race 0.1543
0.0 0.5 1.0 1.5
Mean |SHAP|
(B) XGBoost — SHAP Beeswarm
(blue=low value, red=high value)
Age
BRI
FiberConsumption
ProteinConsumption
CaffeineConsumption
CarbohydrateConsumption
CalorieConsumption
BMI
FatConsumption
Hypertension
FamilyIncome
Race
−4 −2 0
SHAP value (impact on RA probability)
Feature value (norm.)
0.00 0.25 0.50 0.75 1.00
(C) EasyEnsemble — Feature Importance
Age 0.0802
BRI 0.0223
Hypertension 0.0206
FiberConsumption 0.0130
SmokingStatus 0.0082
Gender 0.0073
ProteinConsumption 0.0062
CaffeineConsumption 0.0032
FamilyIncome 0.0019
CarbohydrateConsumption 0.0019
FatConsumption 0.0015
Hyperlipidemia 0.0013
0.000 0.025 0.050 0.075
Mean |SHAP|
(D) Feature Rank Correlation
Spearman ρ = 0.718 (p = 0.000)
Perfect agreement
Age
BRI
Hypertension
FiberConsumption
SmokingStatus
Gender
ProteinConsumption
CaffeineConsumption
FamilyIncome
CarbohydrateConsumption
FatConsumption
Hyperlipidemia
Diabetes
BMI
Race
BRI_Grp
CalorieConsumption
PhysicalActivity
MaritalStatus
EducationLevel
BRI_Trend
DrinkingStatus
EasyEnsemble Feature Rank
0 5 10 15 20
XGBoost Feature Rank
20 10 0